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International Journal of Refractory Metals & Hard Materials 18 (2000) 301–306

International Journal of
REFRACTORY METALS
& HARD MATERIALS

Trends and market perspectives for diamond tools in the construction industry

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Received 26 May 2000; accepted 19 June 2000

Abstract

New materials technologies continue to emerge and, as they develop, they enable both the invention of new products and the displacement of traditional products. Although diamond as a natural material has a long technical history it is still one of the most innovative materials for tools and many other applications. Due to its hardness, abrasion resistance, thermal conductivity and many other favourable properties, diamond as an engineering material has long been pursued by scientists and engineers. New fabrication and processing technologies have caused a "diamond fever" in the tooling industries for metal cutting, as well as drilling or surfacing of construction materials or ultrahard materials.

This paper gives an overview on the current status of diamond technologies, with a special focus on their application in the construction industry, and highlights corresponding market trends. The construction industry is an interesting example where the diamond tool meets exceptionally high requirements regarding performance and lifetime. Modern materials technologies such as powder metallurgy and coating technologies have proved to have the potential to fulfill these requirements.

1. Introduction

Diamond tools are applied in a wide variety of precision machining tools for jet engines, automobiles, electronic parts, for the medical treatment of teeth and eyes, for the manufacturing of tomb stone and for cutting, drilling and surfacing of construction materials. Many of these sophisticated manufacturing tasks can only be tackled by employing diamonds as the key tooling material. Due to the exceptional properties of diamond in line with new fabrication and materials technologies the areas of applications are steadily growing. New fields of interests are wear-resistant coatings on glass or plastics, optical elements or heat sinks for high-performance computers. Diamond coatings especially are expected to grow into a large market [1].

Looking back into history, diamond as a superhard material has long attracted scientists and technicians. Back in 1862, the Swiss engineer J.R. Leschot conceived the idea of making diamond drill bits in an annular ring. Only five years later the first deep hole of 22,860 m was drilled by M.C. Bullock, near Portswill in America [2]. The physical and mechanical properties of rock including its hardness, abrasion resistance and erosive properties impose tough requirements on tools used in the stone and construction industries. Diamond as a natural or synthetic material is able to fulfill those technical demands. Looking at the mechanical properties (see Fig. 1) it becomes obvious that no other ceramic or hard material can substitute diamond in terms of hardness. However, the major drawback is its limited fracture toughness. In order to achieve optimal cutting, sawing or grinding behaviour, the tool design has to take this combination of features into consideration by developing "application adapted" materials concepts. Furthermore, the availability of synthetic diamond was a further obstacle that had to be overcome in the past as natural diamond is a very rare material. The desire to master the secret of diamond synthesis has occupied countless generations of alchemists and scientists. Success was not to come until the middle of the 20th century. During World War II, the American companies General Electric, Norton Company and Carborundum Company entered into an agreement aimed at

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0263-4368/01/$ - see front matter
PII: S0263-4368(00)00034-2

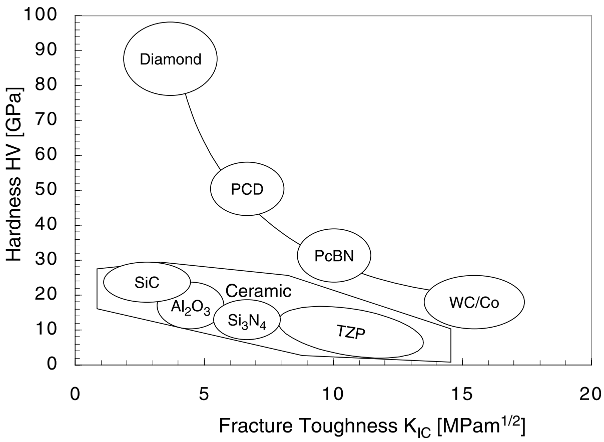

Fig. 1. Mechanical properties of selected hard materials.

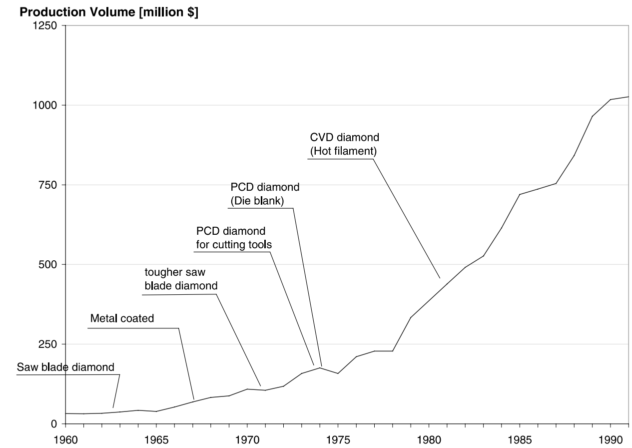

Fig. 3. Commercialization time schedule for superabrasives [16].

synthesizing diamond. Advantage was taken of the work of Nobel prize winner Percy W. Bridgeman, who worked on high pressure physics. By the end of 1957, GE had mastered the process and synthetic diamond became a commercial reality. The basis for the synthesis process is summarized in the famous paper by Berman and Simon [3] which describes the graphite–diamond equilibrium. Temperatures and pressures typically employed under industrial conditions are 1350–2100 K and 4.5–10 GPa (approx. 50,000 times the atmospheric pressure) [4]. Diamond is recovered from the post-reaction synthesis mass after the metal and unreacted graphite have been removed. The original synthesis process has since been modified and optimized in various ways. Modifications aim at the manufacturing of larger dimensions or at new processes, such as the shock wave synthesis or the use of new catalysts [4]. The high-pressure synthesis process can produce large monocrystalline diamonds of up to 5 carats [5]. Fig. 2 shows typical synthetic diamonds ready for application in diamond tools. Since the first commercialization in the sixties the market for superabrasives has steadily grown, as can be seen in Fig. 3. New developments regarding the manufacturing routes for synthetic diamond have fostered the significant market growth. In 1995, the global market for finished end use diamond products including polycrystalline products and thin film coatings has been estimated to reach a volume of \$3,5b [6]. Latest estimates for the market share of diamond tools in the construction industry in 1997 give a market volume of \$1,8b and the demand is still growing.

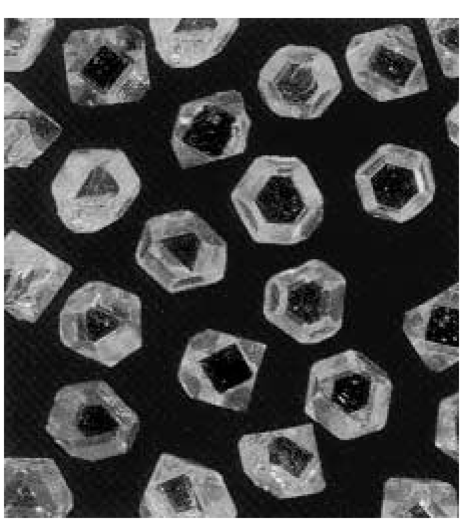

Fig. 2. Synthetic diamond grit for drilling tools, *Courtesy of DeBeers*.

2. High pressure synthesized diamonds in tools for the construction industry

2.1. Single crystalline diamond tools

Stone and construction materials have high abrasion resistance and abrasivity and hence impose extreme

Fig. 4. Diamond drilling tool in operation; *Hilti AG, Schaan*.

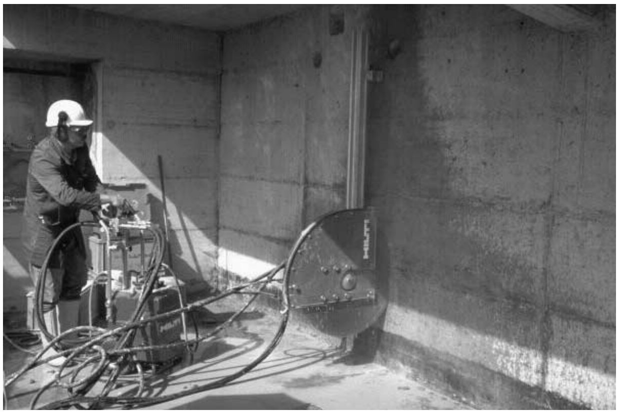

Fig. 5. Diamond wall saw in operation; *Hilti AG, Schaan.*

Fig. 6. Sintered diamond drilling segment; *Hilti AG, Schaan.*

requirements on a cutting, grinding or sawing tool. Consequently, an appropriate tool needs to have an extremely high wear resistance. Furthermore, it is crucial that the tool shows high cutting and penetration rates over extended lifetime. Modern diamond tools have proven that they can fulfill these requirements. In Fig. 4, a high-performance diamond drilling tool is shown. The machine is operated at rotational speeds from 1200 to 3000 rpm at a power of 1700 W. Fig. 5 shows a modern diamond sawing tool. The wall sawing tool is operated at speeds of 850–1270 rpm with sawing disks of up to 1200 mm in diameter.

Both tools have in common that synthetic diamonds fabricated via high-pressure synthesis are employed. The single diamond crystals are embedded in a segment that is subsequently joined to the specific tool body. The segment manufacturing process is done by either a cold pressing and sintering or hot pressing. In cold pressing and sintering, self-bonding powders are used that are formed into tool shapes by pressing in moulds. They are then removed from the moulds to be placed, free standing, in an atmosphere controlled furnace for sintering. In contrast to cold pressing and sintering, the hot pressing process consists of heating a diamond impregnated metal bond mixture under high pressure. Fig. 6 shows a diamond impregnated segment ready for use in a diamond drilling tool. Although the manufacturing costs of metal bond diamond tools are rather high they are almost exclusively used for sawing and drilling of concrete, stone, construction materials or glass.

The crucial materials parameters for impregnated diamond tools regarding performance on construction materials or stone are diamond size and quality, diamond concentration, the tool kinematics and the bond itself. Particle size, quality and distribution, together with the machining parameters have a significant influence on the grit wear mode. The ideal grit breakdown behaviour occurs when the wear progression of the particle contains an optimum balance between wear-flattening and microfracture. This can clearly be seen in Fig. 7. Before the cutting edges of individual crystals in the diamond impregnated segment flatten, new cutting edges are created by micro-chipping. The self-sharpening mechanism depends on the right balance between friction and bit wear. Research tries to understand these complex mechanisms in order to provide the right grit and the right concentration in the segment depending on the cutting job to be done [7–9]. The bonding mechanism between diamond and metal binder determines how the tool operates. With a good bonding in a soft matrix, it operates at high stock removal rates and low contact pressure, but with a short operating life. Harder matrices with poorer bonding, give lower stock removal rates and high contact pressures, but with a longer

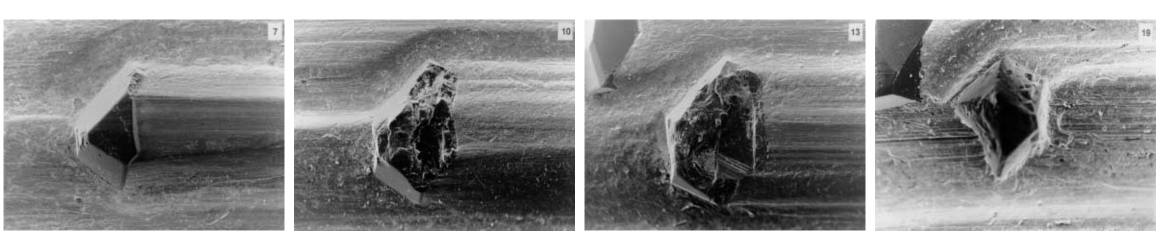

Fig. 7. SEM micrograph of a single diamond in a sintered segment during cutting. The diamond shows cleavage controlled microfracture as part of the self-sharpening mechanism. The last picture illustrates the final pull-out from the matrix.

operating life. Consequently, the segment's operating temperature is higher in the case of harder matrices. This is of course, also strongly influenced by the material to be removed. Hence, industrial suppliers of diamond tools have started to advise the customer in the selection of the most appropriate diamond tool [10]. In order to achieve higher removal rates at longer operating lifes research starts to investigate alternative metal binder systems to the most common cobalt or bronze binders. Cu-systems are one group that is currently being investigated. Alloy modifications are being made in the Cu–Sn, Cu–Zn–Al or Cu–Ti systems. The idea is to increase the bonding strength by inducing an interfacial interaction between binder and diamond and to enhance the abrasion properties. To reduce friction, graphite is sometimes also added to the bond composition.

2.2. Polycrystalline diamond tools

A second concept of employing diamonds in tools for the construction industry is with polycrystalline diamonds. As Fig. 3 shows, polycrystalline tools have gained commercial importance after larger diamond cutting bits with defined cutting edges were asked for in metal and wood cutting, as well as in specific applications in the construction industry. Fig. 8 shows a PCD body with a defined edge structure. The tool has been polished carefully to give a good edge. However, the PCD edges shows marked imperfections which are particularly obvious in the top left part of the picture. In comparison to single crystals, PCD tools have a superior toughness, or resistance to chipping. Hence, PCD tools are employed when deep cuts and large forces are involved. A further advantage of PCD material is that, it is isotropic without the crystal orientation dependent cleavage of single crystal diamond.

The basic tool manufacturing approach is to sinter fine diamond crystals together to form a polycrystalline mass (PCD). In order to fabricate a PCD cutting bit, it is crucial to bond the PCD plate to a substrate. Brazing is a common approach to manufacture high strength PCD tool bodies. Although the processes for brazing diamond are rather common, there are still quite a few drawbacks to overcome. The key areas are to lower the brazing temperature and time and to increase the bonding strength. The brazing interlayer alloy ideally wets both parent materials and a metallurgical interaction is induced. Designing the braze alloy to achieve a high bonding strength means adding a certain amount of reactive agents to the braze alloy. Active alloy components can interact with the diamond to achieve a good wetting and bonding of the braze without deteriorating the diamonds. Regarding process optimization, new developments are focusing on lower melting temperature braze alloys as well as on shorter bonding times, for instance by employing novel techniques such as resistance brazing [11].

Fig. 8. View of the cutting edge of a PCD tool [5].

2.3. Market for diamond tools

In 1995, the market for PCD tools has been estimated to have a volume of \$242 m. The technological developments in the 90's have led to new forecasts that indicate very high growth rates for PCD and polycrystalline cubic boron nitride (PCBN) for the coming years [12,13]. Stone processing, the construction industry and the mining industry have generated a strong demand for impregnated as well as polycrystalline diamond tools. Interesting applications for PCD tools are investigated in rock drilling. A new hybrid bit concept, where thermally stable PCD bits have been embedded in diamond impregnated segments, has made it feasible to cut into medium-to-hard rock formations. This approach is an excellent example, showing that materials engineering solutions combined with deep understanding of the penetration mechanisms has the potential to provide new solution to complex demolition problems [14]. Further, new applications for PCD and diamond tools arise in the wood processing industry. Wooden laminates for masonry flooring are a growing market that impose extraordinary requirements on the tool due to their abrasive alumina coating. Tools with large monocrystalline diamonds have proven to be more reliable than even conventional PCD cutting bits [15].

In general, the market for diamond tools is rapidly growing. Current figures from DeBeers indicate that the demand for diamond abrasives has reached a volume of 700 m carats in 1997. Looking at the diamond consumption for sawing tools in 1997, it is interesting to see that Europe is by far the largest market. 45% of the 270–280 m carats of diamonds for saws produced in 1997 went into the European market. In comparison, North America and Japan both had a saw diamond

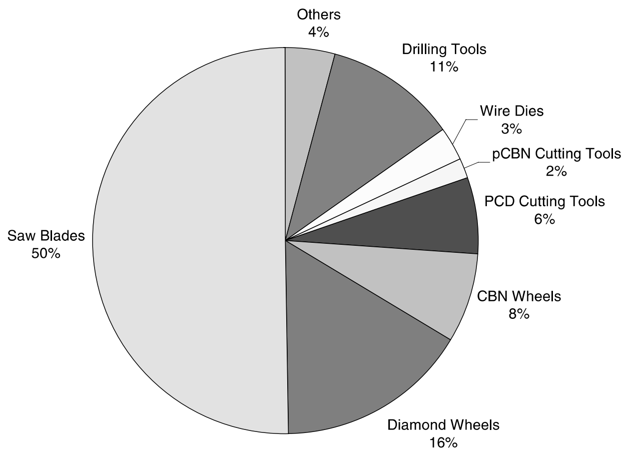

Fig. 9. Applications for superabrasive tools in Europe in 1992 [16].

consumption of 14% each [12]. In Fig. 9 an analysis of the European market for superabrasive products is given [16]. Again the pie chart indicates that saw blades are the largest market segment. Further, strong market segments are identified for PCD and cubic boron nitride (CBN) tools.

3. Low pressure synthesized diamond tools (CVD)

Growing synthetic diamond at high temperatures and high pressures is only one approach for providing high-perfomance diamonds for various applications. With the invention of the low pressure chemical vapour deposition techniques, diamond became available in the form of extended thin films and free-standing plates. The common feature of all diamond CVD techniques is a gas-phase nonequilibrium, i.e., a high supersaturation of atomic hydrogen and of various hydrocarbon radicals. Typical deposition conditions are: 1% methane in hydrogen as source gas, 700–1000°C deposition temperature and a gas pressure in the range of 40–400 Pa. The most common techniques include thermally assisted CVD (hot filament), microwave plasma assisted CVD, deposition in a combustion flame, and arc jet CVD [17–19].

Adhesion of a deposited diamond film to the tool has presented the greatest technological obstacle to commercial introduction of diamond-coated tool inserts. In the early stages of development, delamination was a common occurrence. Thus, an important goal was to develop adhesion methods which permitted the diamond film to wear out rather than spall off. Hardmetal is the most widely used tool insert material. It typically contains cobalt as a binder for the hard tungsten carbide particles. Cobalt, similar to iron, has a high carbon solubility and therefore it is difficult to deposit high quality diamond on an untreated hardmetal tool. This problem can either be tackled by treating the hardmetal inserts to remove cobalt or by coating it with a barrier layer.

Fig. 10 shows the structure of a CVD diamond layer and corresponding commercially available cutting bits are shown in Fig. 11. The micrograph shows that 3-D coatings are possible with a reliable quality. In order to enhance the commercialization of these low pressure diamond synthesis methods, it is crucial to improve the economics of the CVD process. A good overview regarding market perspectives of diamond CVD processes is given in [20] where it is clearly indicated that the deposition rate is the economic bottleneck for CVD diamond films. Current developments are aiming at improving the process economics and achieving higher bonding strengths. Despite high fabrication costs, CVD-diamond coatings on tools are already commercially available and have shown their high potential in fabrication tools for shaping and turning of graphite and metal-matrix composites [21]. Studies in Japan show that diamond coatings on cemented carbide cutting bits have increased the cutting performance of tools used for shaping cylinder heads in the automotive industry by factor of 4. In cutting stone for masonry constructions the lifetime of diamond coated tool has proved to be 5 times higher than the uncoated tool [22]. Current developments are also aiming to deposit free-standing CVD diamond plates that can substitute PCD in the cutting industry or can be utilized in opto-electric devices [23].

A further area of diamond coatings expected to grow significantly, is the engineering application for diamond-like carbon (DLC) coatings. DLC-coatings are deposited via PVD – or CVD-processes in the presence of gaseous or bulk carbon. Owing to their amorphous structure, DLC-coatings are able to incorporate large

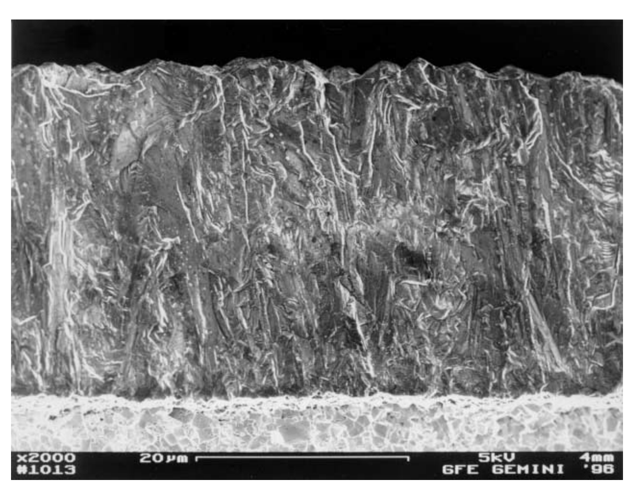

Fig. 10. Coating morphology of a CVD diamond layer, *Courtesy of CemeCon GmbH.*

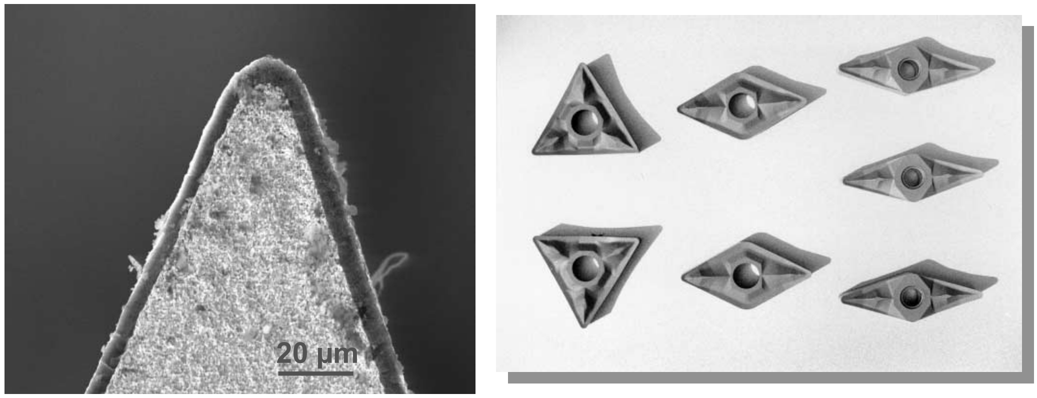

Fig. 11. 3-D diamond coating of cutting bits, *Courtesy of CemeCon GmbH.*

quantities of hydrogen which lead to their alternative name of "a-C:H". Doping the layer with metals enhances the abrasion resistance and consequently a large field of application is in general wear protection. Examples are found in computer hard-disks, aircraft windows, sun-glasses and even razor-blades. Furthermore, DLC coatings have good corrosion resistance and offer low friction which make them also a promising coating material for wood cutting tools. A study in the UK revealed, a UK market volume of EURO 90 m for DLC-coatings for the year 2005. The DLC-coating market for the year 1990 has been estimated to have a volume of EURO 22 m [24,25].

4. Conclusion

Synthetic diamond for high-performance tools is a rapidly growing market. Both principal manufacturing routes, high-pressure and low-pressure synthesis, produce diamonds that open up a large number of tool applications in the construction industry. Academic and applied research continues to show that the full potential of this very old and yet still brand new material has, by far, not yet been exploited.

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